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Efficient Closed-Form Reduction for Banegas' Second Family of Pentanomials

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ABSTRACT In polynomial-basis arithmetic over $\mathbb{F}_{2^m}$, the defining irreducible polynomial determines the cost of modular reduction and hence of field multiplication. Using a greedy XOR-counting heuristic, Banegas identified two families of pentanomials with low reduction cost. The first family has since been analyzed, but the second one, $f_{b,c}(x) = x^{b+2c} + x^{b+c} + x^b + x^c + 1$ with $b > c \geq 1$, was left open because its reduction has a less transparent structure. This paper closes that gap. For $m = b + 2c$, we derive a closed form for the quotient of the polynomial division, written in terms of comb sums of stride $3c$, and turn it into an explicit reduction circuit with at most $3m - c - 3$ XOR gates and depth at most $\lfloor (m-2)/(3c) \rfloor + 4$. A second construction splits the reduction into c independent chains and uses $3m - 3 - N(b, c)$ XOR gates, where $N(b, c) \approx (m - c)/3$; it is shorter whenever $b \geq 4c + 1$. Both constructions approach $8m/3$ gates, at opposite ends of the range of c. Members built from five small base pentanomials split into independent strips; composing strip programs that a SAT solver proves optimal for the base polynomials gives even shorter circuits. We verified both constructions symbolically for all 201,117 members with $m \leq 1100$. An irreducibility census for $5 \leq m \leq 1100$ found irreducible members in 720 degrees, 288 of which have no irreducible trinomial. Wherever both families have irreducible members, in 405 degrees, the reduction bounds of the second family are lower, by 9.7% on average. At the NIST degrees 163 and 571, the best members need 440 and 1,528 XOR gates for the reduction, against 571 and 2,003 for the NIST pentanomials.

INDEX TERMS Binary finite fields, bit-parallel multipliers, irreducible pentanomials, modular reduction, polynomial basis, XOR circuit complexity.

I. INTRODUCTION

FINITE fields play an important role in cryptography and error-correcting codes [1], [2]. Two types are particularly relevant in practical implementations: prime fields $\mathbb{F}_p$, where p is a large prime, and binary extension fields $\mathbb{F}_{2^m} \cong \mathbb{F}_2[x]/(f)$, where f is an irreducible polynomial of degree m. Arithmetic in prime fields relies on multiprecision integer operations, often taking advantage of carefully chosen primes to simplify modular reduction. In binary fields, addition requires only bitwise XOR operations, while multiplication consists of a carry-less polynomial product followed by reduction modulo f.

Several representations are available for binary fields, including polynomial, normal, shifted and Montgomery bases, as well as towers of subfields. In a polynomial basis, the multiplication of two polynomials does not depend on the choice of f, but the subsequent reduction does. The selection of the irreducible polynomial can therefore have a substantial impact on implementation cost, particularly in terms of circuit area, delay and energy consumption. It also affects the number of operations required in bitsliced software implementations.

Although recent standardization efforts in elliptic curve cryptography have favored prime fields, binary fields remain relevant in several cryptographic constructions. NIST SP 800-186 retains prime-field curves, introduces Edwards curves and deprecates the previously standardized binary-field curves [3], [4]. Those binary curves used trinomials of degrees 233 and 409, and pentanomials of degrees 163, 283 and 571.

Outside elliptic curve cryptography, binary fields continue to appear in widely used algorithms and more recent proposals. AES uses $\mathbb{F}_{2^8}$ [5], while GCM performs authentication using a degree-128 pentanomial [6], [7]. Classic McEliece employs fields of degrees 12 and 13 [8], [9], and the post-quantum signature scheme FAEST uses $\mathbb{F}_{2^\lambda}$ [10]. Towers of binary fields also appear in succinct argument systems [11]. Beyond cryptography, Reed-Solomon and BCH codes provide further examples of applications in which efficient binary-field arithmetic is important [12].

The cost of polynomial reduction depends strongly on the number and positions of the nonzero coefficients of f. Irreducible trinomials are especially attractive, since reduction can be implemented using $2m-2$ XOR gates [13], [14]. However, irreducible trinomials do not exist for every degree. In particular, Swan's theorem rules them out when $m \equiv 0 \pmod 8$, as well as for many odd degrees satisfying $m \equiv \pm 3 \pmod 8$ [15]. Pentanomials are a natural alternative in these cases, and considerable effort has been devoted to identifying families that permit efficient reduction or multiplication [16]–[21].

An earlier investigation of this problem was carried out by Banegas in his master's thesis at LabSEC/UFSC [22]. Using a greedy XOR-counting algorithm called *Conta-XOR*, he examined all irreducible pentanomials of degrees 19 and 163 and identified two families with particularly low reduction costs:

$$\begin{aligned} &\text{(first)} && x^{2b+c}+x^{b+c}+x^b+x^c+1,\\ &\text{(second)} && x^{b+2c}+x^{b+c}+x^b+x^c+1. \end{aligned}$$

The first family was subsequently studied by Banegas, Custódio and Panario, who established a reduction cost of $3m-2$ XOR gates, decreasing to $\frac{12}{5}m-1$ when $b=2c$ [21]. The second family, however, was left without a corresponding analysis. Banegas observed that a better understanding of its reduction process was needed and explicitly identified the problem as a direction for future work [22, Sec. 4.5].

The present work addresses this open problem. We show that the second family admits an algebraic structure that makes it possible to describe polynomial reduction in closed form. The central observation is the identity

$$(x^c+1)f = x^{b+3c}+x^b+x^{2c}+1,$$

which provides a sparse multiple of the defining polynomial. Using this identity, we derive an explicit expression for the quotient in polynomial division and obtain a reduction algorithm whose XOR cost and circuit depth can be bounded analytically.

The main contributions of this paper are:

1) We establish structural properties of the second family of pentanomials, including the sparse-multiple identity used throughout the analysis, and show that reduction splits into independent strips modulo a base polynomial G with $f_{b,c}(x) = G(x^{\gcd(b,c)})$. This explains why the members built from five small base polynomials are cheaper, and we give optimal strip programs for them (Section III).
2) We derive a closed-form expression for the quotient $\lfloor D/f \rfloor$ using comb sums with stride $3c$ (Theorem 7).
3) We construct an explicit reduction algorithm for all $b > c \geq 1$, requiring at most $3m-c-3$ XOR gates and having depth at most $\lfloor (m-2)/(3c) \rfloor + 4$. The corresponding multiplier requires $m^2+m-c-2$ XOR gates (Theorem 8, Proposition 9, and Corollary 10).
4) We show that this construction is not optimal and give a second one, based on identities between sums of comb sums, which splits the reduction into c independent chains. It uses exactly $3m-3-N(b,c)$ XOR gates, improves on $3m-c-3$ whenever $b \geq 4c+1$, and approaches $\frac{8}{3}m$ when c is small. For several small members it matches the exact minima, or the best programs, found with a SAT solver (Theorem 13 and Table 5).
5) We develop a word-level formulation with $O(\log(m/c))$ shift-and-XOR operations and a gate-level variant of lower depth (Section VII).
6) We compare the proposed reduction with the greedy Conta-XOR algorithm. For generic members of the family, the closed-form construction is never more expensive, accounts for the counts reported by Banegas and improves them by exactly $2c-b$ XOR gates when $b < 2c$ (Section VIII).
7) We perform an irreducibility census for degrees up to 1100, evaluate both constructions on it, characterize the irreducible special members, and compare the resulting polynomials with those of the first family, with type I and type II pentanomials, with irreducible trinomials and with the NIST pentanomials (Section IX).

The remainder of the paper is organized as follows. Section II introduces the reduction cost model and reviews relevant previous results. Section III examines the second family and establishes its main algebraic properties. The closed-form quotient is derived in Section IV and used to construct the reduction algorithm in Section V. Section VI presents the chain construction. Section VII discusses word-level implementation and a lower-depth circuit variant. Comparisons with Conta-XOR and other polynomial families are presented in Sections VIII and IX, followed by the discussion and conclusions in Section X.

II. BACKGROUND

Let $f = x^m+g$, where $g(0)=1$ and $\deg g < m$. The product of two elements of $\mathbb{F}_{2^m}$ is a polynomial $D = \sum_{j=0}^{2m-2} d_j x^j$, which can be written as

$$\begin{aligned} D &= L + x^m H, \qquad L = \sum_{i=0}^{m-1} d_i x^i,\\ H &= \sum_{s=0}^{m-2} h_s x^s, \qquad h_s = d_{m+s}. \end{aligned} \tag{1}$$

Reduction modulo f produces $R = D \bmod f$. Since reduction is a fixed $\mathbb{F}_2$-linear map, it can be represented by a matrix $R_f \in \mathbb{F}_2^{m\times(2m-1)}$. Column j of this matrix contains the coefficients of $x^j \bmod f$. Thus, $R = R_f(d_0, \ldots, d_{2m-2})^\top$, and each output bit r_i is obtained by XORing the inputs indicated by the corresponding row of R_f.

The implementation cost of this linear map depends on how the XOR operations are organized. A *linear straight-line program* consists of instructions $t \leftarrow u \oplus v$, where the operands are either input variables or previously computed values. Its length is the number of XOR gates, while its *depth* is the maximum number of dependent gates along any computation path. Computing each output independently requires $\mathrm{dXOR}(f) = \sum_i (w_i - 1)$ XOR gates, where w_i denotes the weight of row i of R_f. We refer to this quantity as the *direct count*.

A smaller circuit may be obtained by sharing intermediate sums among different outputs. For example, consider $y_0 = x_1 + x_2 + x_3$, $y_1 = x_1 + x_2 + x_4$, and $y_2 = x_2 + x_3 + x_4$. Computing the three sums independently requires six XOR gates. By first computing $t = x_1 \oplus x_2$, the first two outputs become $y_0 = t \oplus x_3$ and $y_1 = t \oplus x_4$. Together with the two gates required for y_2, this gives a total of five gates.

We denote by $\mathrm{Red}(f)$ the minimum number of XOR gates required by a linear straight-line program implementing R_f. Finding this minimum is NP-hard in general [23], which motivates the use of heuristic methods. Paar's greedy algorithm [12] identifies the pair of operands occurring most frequently among the output expressions, computes that pair once, and replaces its occurrences with a temporary variable. Boyar and Peralta [24] introduced a heuristic that also exploits cancellations arising from the identity $v \oplus v = 0$. Further improvements have been proposed, particularly for the linear layers of block ciphers [25]–[29]. General reduction methods also exist for any low-weight modulus, including a fast reduction used for squaring [30] and a reduction with logarithmic feedback depth for every monic modulus [31]; the present work instead exploits the structure of one family. Exact optimization using SAT solvers is possible for small instances [32], [33], but becomes impractical for the field degrees considered in this work.

Conta-XOR, introduced independently by Banegas [22, Ch. 4], follows a greedy strategy similar to Paar's algorithm. It first constructs the reduction matrix by repeatedly substituting x^m with g and canceling equal terms in each output expression. The algorithm then selects the most frequent pair of operands, replaces it with a temporary variable, and repeats this procedure until no further sharing is possible. Its reported cost is the number of temporary variables introduced plus the XOR operations remaining in the output expressions. We denote this cost by $\mathrm{CX}(f)$.

For the present study, we implemented Conta-XOR independently and reproduced the values reported by Banegas [22]. When several operand pairs have the same frequency, our implementation uses lexicographic ordering to break ties. This makes the results reproducible.

As a greedy method, Conta-XOR does not guarantee a minimum XOR count. A locally favorable choice may prevent more efficient sharing in subsequent steps. Moreover, its pair-substitution strategy does not exploit cancellations between intermediate expressions. Despite these limitations, Conta-XOR was useful in identifying pentanomials with low reduction costs, including the second family studied here. Its output provides an upper bound on $\mathrm{Red}(f)$ rather than an exact minimum. We compare these heuristic counts with those of our closed-form construction in Section VIII.

Several families of irreducible polynomials admit reduction algorithms whose XOR costs are known in closed form. Table 1 summarizes the results relevant to our comparison, including trinomials, equally spaced pentanomials, type I pentanomials [17], [18], [20], general pentanomials with small middle exponents, and Banegas' first family. Type II pentanomials $x^m + x^{n+2} + x^{n+1} + x^n + 1$ [17] are compared in Section IX-C.

For a schoolbook polynomial product, the partial products require m^2 AND gates and $(m-1)^2$ XOR gates. The total XOR count of a multiplier is therefore $(m-1)^2$ plus the cost of modular reduction. For Banegas' first family, this gives $m^2 + m - 1$ XOR gates [21].

III. THE SECOND FAMILY

Definition 1. Let $b > c \geq 1$ be integers and set $m = b + 2c$. The pentanomials

$$f_{b,c}(x) = x^{b+2c} + x^{b+c} + x^b + x^c + 1$$

form *Banegas' second family*. Throughout the paper, $n = m - 2$ denotes the degree of the polynomial H in (1).

The restriction $b > c$, adopted in [22], ensures that the five exponents are distinct. In particular, $c < b < b + c < b + 2c$.

Proposition 2. *Let $f = f_{b,c}$. Then:*

(i) $x^m \equiv (x^b + 1)(x^c + 1) \pmod f$.

(ii) $(x^c+1)f = x^{b+3c}+x^b+x^{2c}+1$. *Consequently,* $x^b(x^{3c}+1) \equiv (x^c + 1)^2 \pmod f$.

(iii) $1 \leq c \leq \lfloor (m-1)/3 \rfloor$ *and* $b \equiv m \pmod 2$.

(iv) If $\gamma = \gcd(b, c) > 1$, *then* $f(x) = G(x^\gamma)$, *where* $G = f_{b/\gamma, c/\gamma}$ *also belongs to the family. If* γ *is even, then* $f = G(x^{\gamma/2})^2$ *and is therefore reducible.*

(v) If $b = 2c$, *then* f *is the equally spaced pentanomial* $x^{4c} + x^{3c} + x^{2c} + x^c + 1$.

Proof. (i) Writing $f = x^m + g$, we have

$$g = x^{b+c} + x^b + x^c + 1 = (x^b + 1)(x^c + 1).$$

Thus $x^m \equiv g \pmod f$.

(ii) Multiplication by $x^c + 1$ gives

$$(x^c + 1)f = x^{b+3c} + x^b + x^{2c} + 1,$$

since the remaining terms cancel in characteristic 2. Reducing this identity modulo f yields

$$x^b(x^{3c} + 1) \equiv x^{2c} + 1 = (x^c + 1)^2 \pmod f.$$

TABLE 1. Reduction costs of families of irreducible polynomials reported in the literature.

Polynomial	Condition	XORs for reduction	Ref.
trinomial $x^m + x^a + 1$	$a \le m/2$	$2m - 2$	[13], [14]
equally spaced $x^{4c} + x^{3c} + x^{2c} + x^c + 1$	$m = 4c$	$\frac{7}{4}m - 1$	[16], [22]
type I pentanomial $x^m + x^{n+1} + x^n + x + 1$	$2 \le n \le m/2 - 1$	$3m - 2$	[18], [21]
pentanomial $x^m + x^{k_3} + x^{k_2} + x^{k_1} + 1$	$k_3 \le m/2$	$4m - 4$	[18]
Banegas' first family $x^{2b+c} + x^{b+c} + x^b + x^c + 1$	$b > c$	$3m - 2$	[21]
the same, with $b = 2c$	$m = 5c$	$\frac{12}{5}m - 1$	[21]

(iii) Since $b \ge c+1$, we have $m = b + 2c \ge 3c + 1$, which gives $c \le \lfloor (m-1)/3 \rfloor$. Also, $m - b = 2c$, so b and m have the same parity.

(iv) Let $b = \gamma b'$ and $c = \gamma c'$. Then $b' > c' \ge 1$ and

$$f(x) = f_{b',c'}(x^\gamma) = G(x^\gamma).$$

If γ is even, the characteristic-2 identity $G(x^\gamma) = G(x^{\gamma/2})^2$ shows that f is reducible.

(v) Substituting $b = 2c$ into the definition gives the stated pentanomial. □

The identity in Proposition 2(ii) is central to the reduction method developed in this paper. To obtain it, write

$$f = x^b(x^{2c} + x^c + 1) + (x^c + 1).$$

Since

$$(x^c + 1)(x^{2c} + x^c + 1) = x^{3c} + 1$$

over $\mathbb{F}_2$, multiplication by $x^c + 1$ produces a multiple of f containing only four nonzero terms. The resulting congruence can be expressed as

$$x^{s+b+3c} \equiv x^{s+b} + x^{s+2c} + x^s \pmod{f}, \qquad s \ge 0.$$

This relation allows a high-degree term to be replaced by three lower-degree terms, with a reduction step of $3c$ in the leading exponent.

For example, taking $b = 11$ and $c = 2$ gives

$$f = x^{15} + x^{13} + x^{11} + x^2 + 1$$

and

$$(x^2 + 1)f = x^{17} + x^{11} + x^4 + 1.$$

Therefore,

$$x^{17} \equiv x^{11} + x^4 + 1 \pmod{f}.$$

A similar identity holds for Banegas' first family:

$$(x^b + 1)f = x^{3b+c} + x^{2b} + x^c + 1.$$

Both families are regions of one expression: with $F_{u,v} = x^{2u+v} + x^{u+v} + x^u + x^v + 1$, the first family is $F_{u,v}$ with $u > v$ and the second is $f_{b,c} = F_{c,b}$, with $u < v$. Exchanging the parameters does not carry the earlier analysis over, because the spacing between the leading exponents changes. For the first family, the corresponding stride is $3b$, which exceeds the field degree $m = 2b + c$. For the second family, the stride is $3c < m$. Consequently, reduction in the second family may involve several successive applications of the same relation, particularly when b is large relative to c.

This distinction becomes explicit in the closed-form quotient derived in Section IV. In the first family, the corresponding comb sums contain at most one coefficient, whereas in the second family they may contain several. The parameter c therefore influences both the reduction cost and the depth of the resulting circuit.

Definition 3 (Special and generic members)**.** Let $\gamma = \gcd(b, c)$, and write $b = \gamma b'$ and $c = \gamma c'$. Then $f_{b,c}(x) = G(x^\gamma)$, where the *base polynomial* $G = f_{b',c'}$ has degree $\kappa = b' + 2c'$ (Proposition 2(iv)). A member of the family is called *special* if its base polynomial is one of

$$f_{2,1}, \quad f_{3,1}, \quad f_{4,1}, \quad f_{5,1}, \quad f_{3,2},$$

that is, if $b/c \in \{\frac{3}{2}, 2, 3, 4, 5\}$, or equivalently if $\deg G \le 7$. All other members are called *generic*.

The classification depends only on the ratio b/c, and not on the degree m. Each of the five base polynomials produces special members $G(x^\gamma)$ in every degree $m = \gamma\kappa$. For special members, $c' \le 2$ and therefore $\gamma \ge c/2$. The equally spaced case $b = 2c$ corresponds to the base polynomial $f_{2,1} = x^4 + x^3 + x^2 + x + 1$.

The reason for singling out these five base polynomials is structural. When $\gamma > 1$, reduction modulo f splits into independent reductions modulo G, and Algorithm 1 respects this splitting.

Proposition 4 (Strip decomposition)**.** *Let $f = f_{b,c} = G(x^\gamma)$ with $\gamma = \gcd(b, c)$, $G = f_{b',c'}$ and $\kappa = \deg G$, so that $m = \gamma\kappa$. For $0 \le \rho < \gamma$, call the set of indices congruent to ρ modulo γ the* strip *ρ.*

(i) Reduction modulo f splits into γ independent reductions modulo G, one per strip. Strips $0, \ldots, \gamma - 2$ each reduce a polynomial of degree at most $2\kappa - 1$, and strip $\gamma - 1$ reduces a polynomial of degree at most $2\kappa - 2$. Consequently,

$$\mathrm{Red}(f) \le (\gamma - 1)\, R_1(G) + R_2(G),$$

where $R_1(G)$ and $R_2(G)$ are the lengths of any programs for these two reductions modulo G.

(ii) Every gate of Algorithm 1 combines operands from a single strip. On strip $\gamma - 1$, after the relabeling $i \mapsto (i - \gamma + 1)/\gamma$, it coincides with Algorithm 1 for G and

uses $3\kappa - c' - 3$ gates. On each of the other strips it uses $3\kappa - c'$ gates.

Proof. (i) Write $D = \sum_{\rho=0}^{\gamma-1} x^\rho D_\rho(x^\gamma)$. Since f involves only powers of x^γ,

$$D \bmod f = \sum_{\rho=0}^{\gamma-1} x^\rho \big(D_\rho \bmod G\big)(x^\gamma),$$

because every term on the right has degree less than m. The polynomial D has degree at most $2m-2 = 2\gamma\kappa-2$. Counting the indices $j \le 2\gamma\kappa - 2$ with $j \equiv \rho \pmod\gamma$ shows that D_ρ has degree at most $2\kappa - 1$ for $\rho \le \gamma - 2$ and at most $2\kappa - 2$ for $\rho = \gamma - 1$.

(ii) All index shifts used in Algorithm 1 are c, b, $b + c$ and $3c$, which are multiples of γ, and the high coefficient h_s is the input d_{m+s} with $m \equiv 0 \pmod\gamma$. Hence every gate stays within one strip. On strip $\gamma - 1$, write each index as $\gamma - 1 + \gamma i'$. A direct check shows that every range condition of Algorithm 1 for (b, c) becomes the corresponding condition for (b', c'); for example, $t + 3c \le n$ becomes $t' + 3c' \le \kappa - 2$, and the condition $u \le c - 2$ on the index u of U_u becomes $u' \le c' - 2$. The program on this strip is therefore Algorithm 1 for G, with $3\kappa - c' - 3$ gates. The same check, with indices $\rho + \gamma i'$, shows that strips $0, \dots, \gamma - 2$ run identical programs up to relabeling. Since Algorithm 1 has exactly $3m - c - 3 = \gamma(3\kappa - c') - 3$ gates (proof of Theorem 8), each of these strips uses $3\kappa - c'$ gates. □

Proposition 4(ii) says that the closed form for f is the closed form for G applied once per strip. For a program that respects this decomposition, any saving over $3m - c - 3$ must therefore come from a strip, that is, from a reduction modulo G that is shorter than the closed form. We do not claim that $\mathrm{Red}(f)$ is additive over the strips: a circuit that mixes strips could in principle be shorter, and optimality of the strip programs does not by itself imply optimality of their composition. Writing

$$\delta_1(G) = (3\kappa - c') - R_1(G),$$
$$\delta_2(G) = (3\kappa - c' - 3) - R_2(G),$$

Proposition 4(i) gives

$$\mathrm{Red}(f) \le 3m - c - 3 - (\gamma - 1)\,\delta_1(G) - \delta_2(G). \qquad (2)$$

The saving grows linearly with γ, and hence with m, along each ratio b/c.

Table 2 lists the shortest strip programs for the five base polynomials. They were found by greedy pair substitution with lexicographic and randomized tie-breaking and, for $f_{4,1}$ and $f_{5,1}$, by an exact search with a SAT solver (Section VIII). The SAT solver also proves that all values in the table are optimal. Each program was composed over the strips and checked symbolically against the reduction matrix of f for all γ with $m \le 1100$. For $f_{2,1}$ the bound is $7\gamma - 1 = \frac{7}{4}m - 1$, the known cost of the equally spaced pentanomial (Table 1); for $f_{3,1}$ it is $12\gamma - 2 = \frac{12}{5}m - 2$, and for $f_{4,1}$ it is $15\gamma - 1 = \frac{5}{2}m - 1$.

TABLE 2. Optimal strip programs for the five base polynomials of Definition 3, compared with the closed form, and the resulting bound (2) for $f = G(x^\gamma)$. Optimality of every value of $R_1(G)$ and $R_2(G)$ was proved with a SAT solver. The base polynomials $f_{5,1}$ and $f_{3,2}$, and all their stretched copies, are reducible (Proposition 20); their programs reduce in a quotient ring, not in a field.

G	κ	$R_2(G)$	$3\kappa - c' - 3$	$R_1(G)$	$3\kappa - c'$	bound for f
$f_{2,1}$	4	6	8	7	11	$\frac{7}{4}m - 1$
$f_{3,1}$	5	10	11	12	14	$\frac{12}{5}m - 2$
$f_{4,1}$	6	14	14	15	17	$\frac{5}{2}m - 1$
$f_{5,1}$	7	16	17	18	20	$\frac{18}{7}m - 2$
$f_{3,2}$	7	16	16	18	19	$\frac{18}{7}m - 2$

We applied the greedy search, with 200 randomized runs per reduction, to all 163 reduced pairs (b', c') with $4 \le \kappa \le 40$. Only the five base polynomials of Definition 3 admit a strip program shorter than the closed form that these heuristics can find. For every other pair, including the degree-8 pair $(6, 1)$, the best heuristic program matches or exceeds the closed form. This is the reason for the threshold $\deg G \le 7$. When G is very small, the strip reduction involves only a few outputs and at most seven quotient coefficients, and there is room for groupings of the same sums that are shorter than the uniform structure of Algorithm 1. In the extreme case $G = f_{2,1}$, which divides $x^5 + 1$, reduction modulo G is little more than a cyclic shift, and Algorithm 1 even computes some values twice; for the other four base polynomials, all the values computed by Algorithm 1 are distinct, and the savings come from different groupings.

The classification should be read as a statement about what pair-substitution heuristics find, not about $\mathrm{Red}(G)$. Generic members can also beat the closed form. The chain construction of Section VI does so for every member with $b \ge 4c + 1$, and an exact search gives $\mathrm{Red}(f_{6,1}) = 19$ and $\mathrm{Red}(f_{5,2}) = 21$, one gate below the closed form (Section VIII). What sets the five base polynomials apart is that their savings are found by simple heuristics and that we have provably optimal strip programs for them.

IV. A CLOSED FORM FOR THE QUOTIENT

Polynomial division can be expressed in terms of the coefficients of the dividend, quotient and remainder.

Lemma 5 (Division identities)**.** *Let $f = x^m + \sum_{e\in S} x^e + 1$, with $S \subseteq \{1, \dots, m-1\}$, and let $Q = \lfloor D/f \rfloor = \sum_{j=0}^{n} q_j x^j$. Then*

$$h_j = q_j + \sum_{e\in S} q_{j+m-e}, \qquad 0 \le j \le n, \qquad (3)$$

$$r_i = d_i + \sum_{e\in S\cup\{0\}} q_{i-e}, \qquad 0 \le i \le m-1, \qquad (4)$$

where $q_k = 0$ for $k < 0$ or $k > n$.

Proof. Write $D = Qf + R$, with $\deg R < m$. Comparing the coefficients of x^{m+j} gives (3), since R has no terms of degree m or higher. Similarly, comparing the coefficients of

x^i for $i < m$ gives (4), since $x^m Q$ does not contribute to these terms. □

Equation (3) is triangular: each q_j depends only on h_j and quotient coefficients with indices greater than j. It can therefore be solved in descending order, starting with $q_n = h_n$. Once the quotient is known, (4) determines the remainder. In particular, the quotient depends only on the high part H of the dividend.

For example, consider $f_{3,1} = x^5 + x^4 + x^3 + x + 1$. Here $m = 5$, $S = \{1, 3, 4\}$ and $n = 3$. Equation (3) becomes

$$h_j = q_j + q_{j+1} + q_{j+2} + q_{j+4}.$$

Solving from the highest index yields

$$\begin{aligned} q_3 &= h_3, & q_2 &= h_2 + h_3, \\ q_1 &= h_1 + h_2, & q_0 &= h_0 + h_1 + h_3. \end{aligned}$$

For the second family, $S = \{c, b, b + c\}$ and $m - S = \{b + c, 2c, c\}$. Thus, (3) takes the form

$$h_j = q_j + q_{j+c} + q_{j+2c} + q_{j+b+c}. \tag{5}$$

Direct evaluation of this recurrence requires successive computation of the quotient coefficients, introducing dependencies between them. A closed-form solution avoids these dependencies and provides a more suitable starting point for circuit optimization.

To derive such a solution, consider the polynomial $1 + y^c + y^{2c}$ over $\mathbb{F}_2$. Its formal power-series inverse satisfies

$$\frac{1}{1 + y^c + y^{2c}} = \frac{1 + y^c}{1 + y^{3c}} = (1 + y^c)(1 + y^{3c} + y^{6c} + \cdots).$$

This identity suggests grouping coefficients whose indices differ by multiples of $3c$. The additional term y^{b+c} in (5) will be accounted for separately.

Definition 6 (Comb sums)**.** For $t \geq 0$, define

$$C_t = \sum_{\ell \geq 0,\ t+3c\ell \leq n} h_{t+3c\ell}.$$

Thus, C_t is the sum of the high coefficients with indices congruent to t modulo $3c$ and not smaller than t (see Fig. 1). In particular, $C_t = 0$ for $t > n$, and

$$C_t = h_t + C_{t+3c}.$$

The coefficients are partitioned into $3c$ residue classes, each forming a sequence with stride $3c$. Within each sequence, the comb sums can be computed recursively, reusing previously obtained values.

For instance, if $b = 7$ and $c = 1$, then $m = 9$, $n = 7$ and the stride is 3. The index sequences are $\{0, 3, 6\}$, $\{1, 4, 7\}$ and $\{2, 5\}$. Some corresponding comb sums are

$$\begin{aligned} C_6 &= h_6, \\ C_3 &= h_3 + h_6, \\ C_0 &= h_0 + h_3 + h_6, \\ C_1 &= h_1 + h_4 + h_7, \\ C_2 &= h_2 + h_5. \end{aligned}$$

Theorem 7 (Closed-form quotient)**.** *For $f = f_{b,c}$ and $0 \leq j \leq n$,*

$$q_j = C_j + C_{j+c} + h_{j+b+c}, \tag{6}$$

where $h_s = 0$ for $s > n$.

Consequently, the polynomial $U = (1 + x^c)Q$, whose coefficients are $U_k = q_k + q_{k-c}$, satisfies

$$U_k = \begin{cases} C_k + C_{k+c} + h_{k+b+c}, & 0 \leq k < c, \\ C_{k-c} + C_{k+c} + h_{k+b}, & k \geq c, \end{cases} \tag{7}$$

and the remainder is

$$r_i = d_i + U_i + U_{i-b} \qquad (0 \leq i \leq m-1;\ U_{i-b} = 0 \text{ if } i < b). \tag{8}$$

Proof. Let $\tilde{q}_j$ denote the right-hand side of (6), extended to all $j \geq 0$. By definition, $\tilde{q}_j = 0$ for $j > n$.

We verify that $\tilde{q}_j$ satisfies (5). Substitution gives

$$\begin{aligned} &\tilde{q}_j + \tilde{q}_{j+c} + \tilde{q}_{j+2c} + \tilde{q}_{j+b+c} \\ &\quad = (C_j + C_{j+c} + h_{j+b+c}) \\ &\qquad + (C_{j+c} + C_{j+2c} + h_{j+b+2c}) \\ &\qquad + (C_{j+2c} + C_{j+3c} + h_{j+b+3c}) \\ &\qquad + (C_{j+b+c} + C_{j+b+2c} + h_{j+2b+2c}). \end{aligned}$$

Since $b + 2c = m > n$, the terms h_{j+b+2c}, h_{j+b+3c}, $h_{j+2b+2c}$ and C_{j+b+2c} vanish. The terms C_{j+c} and C_{j+2c} each occur twice and therefore cancel over $\mathbb{F}_2$.

Furthermore, $C_{j+b+c} = h_{j+b+c}$, since $j + b + 4c > n$. These two terms also cancel, leaving

$$C_j + C_{j+3c} = h_j.$$

Hence $\tilde{q}_j$ satisfies (5). As the recurrence is triangular and has a unique solution with $q_j = 0$ for $j > n$, we obtain $\tilde{q}_j = q_j$, proving (6).

For (7), consider first $0 \leq k < c$. Since $q_{k-c} = 0$, we have $U_k = q_k$, giving the first case.

For $k \geq c$,

$$\begin{aligned} U_k &= q_k + q_{k-c} \\ &= (C_k + C_{k+c} + h_{k+b+c}) \\ &\quad + (C_{k-c} + C_k + h_{k+b}). \end{aligned}$$

The two occurrences of C_k cancel. Moreover, $k + b + c \geq b + 2c = m > n$, so $h_{k+b+c} = 0$. Therefore,

$$U_k = C_{k-c} + C_{k+c} + h_{k+b}.$$

Finally, substituting $S = \{c, b, b + c\}$ into (4) gives

$$\begin{aligned} r_i &= d_i + q_i + q_{i-c} + q_{i-b} + q_{i-b-c} \\ &= d_i + U_i + U_{i-b}, \end{aligned}$$

which proves (8). □

The closed-form expression separates the computation into comb sums, quotient coefficients and remainder coefficients.

It also has a compact reading in terms of the reciprocal polynomial

$$f^*(y) = y^m f(y^{-1}) = 1 + y^c + y^{2c} + y^{b+c} + y^m.$$

Write $P^*(y) = y^k P(y^{-1})$ for a polynomial P of degree at most k, with $k = n$ for H and Q. Reversing $D = Qf + R$ shows that $Q^* \equiv H^* (f^*)^{-1} \pmod{y^{m-1}}$, and

$$(f^*)^{-1} \equiv \frac{1 + y^c}{1 + y^{3c}} + y^{b+c} \pmod{y^{m-1}}. \tag{9}$$

Indeed, with $A = 1 + y^c + y^{2c}$ and $T = y^{b+c}$, the term y^m vanishes modulo y^{m-1}, and T^2 does too because $2(b + c) \geq m - 1$, so $(A + T)^{-1} \equiv A^{-1} + TA^{-2}$. Over $\mathbb{F}_2$, $A^2 = 1 + y^{2c} + y^{4c}$, hence $A^{-2} \equiv 1 \pmod{y^{2c}}$ and $TA^{-2} \equiv T$ because $b + 3c \geq m - 1$; finally $A^{-1} = (1 + y^c)/(1 + y^{3c})$. The rational term of (9) produces the periodic pattern of stride $3c$, that is, $C_j + C_{j+c}$ in (6), and the monomial y^{b+c} produces the isolated correction h_{j+b+c}. Inversion of power series is a general tool; what is specific to the second family is that the inverse collapses to this short form, and that the form translates into circuits with shared intermediate sums.

For reduction, however, it is not necessary to compute Q explicitly. By Proposition 2(i),

$$f = x^m + (x^b + 1)(x^c + 1).$$

Consequently,

$$R = D - Qf \equiv L + (x^b + 1)(x^c + 1)Q \pmod{x^m}.$$

Writing $U = (x^c + 1)Q$ gives (8) directly.

This formulation also reduces the number of intermediate operations. In $U_k = q_k + q_{k-c}$, the common comb sum C_k cancels, leaving two comb sums and at most one additional high coefficient.

Figure 1 illustrates this cancellation for $b = 11$ and $c = 2$. In this case,

$$q_2 = C_2 + C_4, \qquad q_0 = C_0 + C_2 + h_{13},$$

and therefore

$$U_2 = q_2 + q_0 = C_0 + C_4 + h_{13}.$$

V. THE CLOSED-FORM REDUCTION PROGRAM

The closed-form expressions of Theorem 7 allow intermediate results to be shared across the reduction circuit. Each comb sum C_t is computed once and reused in the expressions for U_k. These values, in turn, contribute to both r_k and r_{k+b} whenever the corresponding indices are within range.

Additional sharing is possible for $0 \leq i \leq c - 2$. In this range, the expression $C_i + h_{i+b+c}$ occurs in both U_i and U_{i+c}, since $h_{(i+c)+b} = h_{i+b+c}$. We therefore introduce the intermediate values $Z_i = C_i \oplus h_{i+b+c}$. The resulting reduction procedure is given in Algorithm 1.

The algorithm consists of four stages: computing the comb sums, forming the shared pairs, evaluating the coefficients U_k, and constructing the remainder. Each stage uses values obtained in the preceding stages. All conditional branches depend only on b and c, so fixed parameters determine a straight-line program of XOR operations.

Algorithm 1: Reduction modulo $f_{b,c}$, with $m = b+2c$, $n = m - 2$ and $h_s = d_{m+s}$.

Input : $d_0, \ldots, d_{2m-2}$
Output : $r_0, \ldots, r_{m-1}$ with $\sum r_i x^i = D \bmod f_{b,c}$

for $t \leftarrow n$ **to** 0 **do** // comb sums
  if $t + 3c \leq n$ **then** $C_t \leftarrow h_t \oplus C_{t+3c}$;
  else $C_t \leftarrow h_t$;
for $i \leftarrow 0$ **to** $c - 2$ **do** // shared pairs
  $Z_i \leftarrow C_i \oplus h_{i+b+c}$
for $k \leftarrow 0$ **to** $m - 1$ **do** // values U_k
  if $k \leq c - 2$ **then**
    $U_k \leftarrow Z_k \oplus C_{k+c}$
  else if $k = c - 1$ **then**
    $U_k \leftarrow C_{c-1} \oplus C_{2c-1}$
  else if $k \leq 2c - 2$ **then**
    $U_k \leftarrow Z_{k-c} \oplus C_{k+c}$
  else if $k \leq b + c - 2$ **then**
    $U_k \leftarrow C_{k-c} \oplus C_{k+c}$
  else $U_k \leftarrow C_{k-c}$;
for $i \leftarrow 0$ **to** $m - 1$ **do** // outputs
  if $i \geq b$ **then** $r_i \leftarrow (d_i \oplus U_i) \oplus U_{i-b}$;
  else $r_i \leftarrow d_i \oplus U_i$;

The comb sums are evaluated in descending order because C_t depends on C_{t+3c}. When $t + 3c > n$, the sum contains only h_t and requires no XOR gate.

The computation of U_k is divided into five ranges according to which terms of (7) remain nonzero. The shared values Z_i are used in the first and third ranges. At $k = c - 1$, the additional coefficient h_{m-1} vanishes. For $k \geq 2c - 1$, the term h_{k+b} is zero, and for $k \geq b + c - 1$, C_{k+c} also vanishes. In the final range, $U_k = C_{k-c}$ requires no additional XOR operation.

The remainder coefficients are then obtained from (8). Each output contains $d_i \oplus U_i$, with an additional term U_{i-b} when $i \geq b$.

Theorem 8 (Main result)**.** *For all $b > c \geq 1$, Algorithm 1 computes $D \bmod f_{b,c}$ using at most $3m - c - 3$ XOR gates. Consequently,*

$$\mathrm{Red}(f_{b,c}) \leq 3m - c - 3.$$

Proof. Correctness. The first loop computes the comb sums C_t according to their definition.

For $0 \leq k \leq c - 2$, the shared value Z_k gives

$$Z_k + C_{k+c} = C_k + C_{k+c} + h_{k+b+c} = U_k,$$

by (7). At $k = c - 1$, the term $h_{k+b+c} = h_{m-1}$ vanishes.

For $c \leq k \leq 2c - 2$, we have

$$Z_{k-c} + C_{k+c} = C_{k-c} + h_{k+b} + C_{k+c} = U_k.$$

When $2c - 1 \leq k \leq b + c - 2$, the coefficient h_{k+b} is zero, leaving $U_k = C_{k-c} + C_{k+c}$. Finally, for $k \geq b + c - 1$,

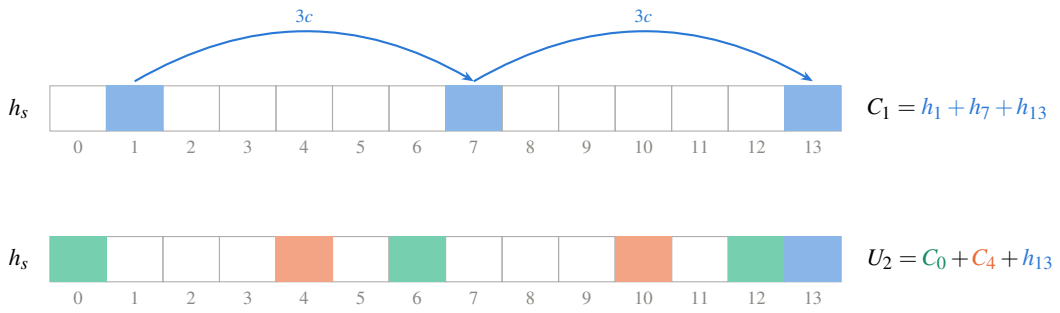


FIGURE 1. Comb sums and the closed-form expression (7) for $b = 11$, $c = 2$ ($m = 15$, $n = 13$, stride $3c = 6$). Top: the comb sum C_1, formed by coefficients at intervals of 6. Bottom: $U_2 = C_0 + C_4 + h_{13}$, comprising two comb sums (green and orange) and one additional high coefficient (blue).

$C_{k+c} = 0$, so $U_k = C_{k-c}$. Thus, every branch computes the corresponding expression in (7).

The final loop implements (8) and therefore produces the coefficients of $D \bmod f_{b,c}$.

XOR count. The comb sums require one XOR for each index satisfying $t + 3c \le n$. Since $n = b + 2c - 2$ and $b > c$, their total cost is

$$n - 3c + 1 = b - c - 1.$$

The shared pairs Z_i require $c - 1$ XOR gates. Computing the values U_k requires one XOR for each $0 \le k \le b + c - 2$, giving $b + c - 1$ gates. No additional gates are needed for the remaining values.

The output stage requires m XOR gates to compute $d_i \oplus U_i$, and another $m - b = 2c$ gates for the outputs containing U_{i-b}.

Adding these contributions gives

$$\begin{aligned}(b - c - 1) + (c - 1) + (b + c - 1) + m + 2c \\ = 2b + 3c - 3 + m \\ = 3m - c - 3.\end{aligned}$$

This establishes the stated upper bound. The count assumes that all operands are distinct. If some of them coincide or cancel, the program becomes shorter, never longer. In our verification up to $m = 1100$, this happens only for the equally spaced members $b = 2c$. □

We verified Algorithm 1 symbolically for all 201 117 members of the family with $m \le 1100$: in every case, the program computes $D \bmod f_{b,c}$ exactly, as a linear map on the $2m - 1$ input coefficients, and uses exactly $3m - c - 3$ XOR gates.

The bound corresponds to fewer than three XOR gates per output bit. For a fixed degree m, increasing c decreases the bound, subject to $b > c$. Members with small c are the most expensive under this bound; Section VI gives a construction that favors them. The implications for polynomial selection are discussed in Section IX.

Two parameter choices illustrate the gate count. For $b = 6$ and $c = 5$, the four stages require $0 + 4 + 10 + 26 = 40$ gates, matching $3 \cdot 16 - 5 - 3 = 40$. In this case, $3c = 15 > n = 14$, so the comb sums require no XOR operations.

For $b = 11$ and $c = 2$, the respective costs are $8 + 1 + 12 + 19 = 40$, again matching $3 \cdot 15 - 2 - 3 = 40$. Here the comb sums account for eight gates, reflecting the smaller stride $3c = 6$.

Proposition 9 (Depth)**.** *The XOR depth of Algorithm 1 is at most*

$$\left\lfloor \frac{n}{3c} \right\rfloor + 4.$$

In particular, it is at most 5 *when* $b \le 4c + 1$.

Proof. Each comb sum C_t is computed through a chain of $\lfloor (n - t)/(3c) \rfloor$ XOR gates. Hence the maximum depth of this stage is

$$\theta = \left\lfloor \frac{n}{3c} \right\rfloor.$$

The shared values Z_i have depth at most $\theta + 1$, and the values U_k have depth at most $\theta + 2$. The final output stage adds at most two XOR levels, giving a maximum depth of $\theta + 4$.

If $b \le 4c + 1$, then $n = b + 2c - 2 < 6c$, so $\theta \le 1$. The depth is therefore at most 5. □

For the 14 748 parameter pairs with $7 \le m < 300$, the measured depth equals $\lfloor n/(3c) \rfloor + 3$ in 97% of the cases. When b is large relative to c, the longer comb-sum chains become the main source of circuit depth.

A balanced-tree implementation of the comb sums reduces the depth to $O(\log(n/c))$, at the expense of additional XOR gates (Proposition 17).

Corollary 10 (Multiplier)**.** *A schoolbook multiplier for* $\mathbb{F}_2[x]/(f_{b,c})$ *requires* m^2 *AND gates and at most*

$$(m - 1)^2 + 3m - c - 3 = m^2 + m - c - 2$$

XOR gates. This is $c + 1$ *fewer XOR gates than the corresponding bound for Banegas' first family (Table 1).*

Example 11. Consider the degree-16 pentanomial

$$f_{6,5} = x^{16} + x^{11} + x^6 + x^5 + 1.$$

It is irreducible, whereas no irreducible trinomial exists in degree $16 \equiv 0 \pmod 8$.

Here $n = 14$ and $3c = 15 > n$, so every comb sum consists of a single coefficient, $C_t = h_t$. The reduction circuit

therefore benefits from sharing the pairs Z_i and the values U_k, without requiring XOR gates for the comb sums.

Table 3 gives the resulting program. It uses 40 XOR gates, compared with a direct count of $\mathrm{dXOR}(f) = 54$ and a Conta-XOR count of $\mathrm{CX}(f) = 44$.

VI. A SHORTER PROGRAM FROM THE CHAIN STRUCTURE

Algorithm 1 is a direct transcription of the closed form, and it is not optimal. Exact minima computed with a SAT solver (Table 5) lie below $3m-c-3$ for every small generic member we could solve. In this section we show where the slack comes from and remove most of it for $b \geq 4c+1$.

In the middle range of (7), Algorithm 1 spends about two gates per value U_k: one to extend a comb sum and one to add two comb sums. The comb sums themselves, however, are never needed. Only the sums

$$E_t = C_t + C_{t+c}, \qquad t \geq 0,$$
$$P_k = C_{k-c} + C_{k+c}, \qquad k \geq c$$

appear in (7): $U_k = E_k + h_{k+b+c}$ for $k < c$, and $U_k = P_k$ for $2c-1 \leq k \leq b+c-2$. These sums can be obtained from each other directly.

Lemma 12 (Chain identities)**.** *For all $t \geq 0$ and $k \geq c$,*

(i) $E_t = h_t + P_{t+2c}$;
(ii) $P_k = h_{k-c} + E_{k+c}$;
(iii) $P_k = E_{k-c} + E_k$.

Moreover, $E_t = h_t$ for $t > n-c$, and $P_k = h_{k-c}$ for $k > n-c$.

Proof. (i) Since $C_t = h_t + C_{t+3c}$, we get $E_t = h_t + C_{t+c} + C_{t+3c} = h_t + P_{t+2c}$. (ii) Similarly, $C_{k-c} = h_{k-c} + C_{k+2c}$ gives $P_k = h_{k-c} + C_{k+c} + C_{k+2c} = h_{k-c} + E_{k+c}$. (iii) The two occurrences of C_k cancel. If $t > n-c$, then $C_{t+c} = 0$ and, since $t+3c > n$, $C_t = h_t$. If $k > n-c$, then $C_{k+c} = 0$ and, since $k+2c > n$, $C_{k-c} = h_{k-c}$. □

Identities (i) and (ii) cost one gate each and alternate between the two kinds of sums: starting from the top, P_k gives E_{k-2c}, which gives P_{k-3c}, and so on. Such a *strand* produces one value E and one value P every $3c$ positions, at one gate each, and needs no comb sum. All shifts are multiples of c, so the computation splits into c independent *chains*, one for each residue class of the index modulo c. Within a chain, consecutive indices differ by c, and we group them in blocks of three. Two strands per chain provide E at two of every three positions and P at two of every three positions. The remaining value P comes from (iii), at one more gate. The cost is therefore five gates per three positions, instead of the six gates of Algorithm 1.

To state precisely which values are computed, let

$$\sigma(t) = \left\lfloor \frac{t+1}{c} \right\rfloor \bmod 3.$$

Since $\sigma(t+c) \equiv \sigma(t)+1 \pmod 3$, the value of σ advances by one along each chain. The program, given in Algorithm 2, computes E_t when $\sigma(t) \neq 2$, obtains P_k from a strand when $\sigma(k) \neq 1$, and uses (iii) when $\sigma(k) = 1$. Values with index above $n-c$ are taken from the last part of Lemma 12 and cost nothing. The values U_k with $c \leq k \leq 2c-2$ use $U_k = U_{k-c} + E_k$, which replaces the shared pairs Z_i of Algorithm 1.

Algorithm 2: Chain construction for reduction modulo $f_{b,c}$, with $m = b+2c$, $n = m-2$, $h_s = d_{m+s}$, and $E_t = h_t$, $P_k = h_{k-c}$ for indices above $n-c$.

Input : $d_0, \ldots, d_{2m-2}$
Output : $r_0, \ldots, r_{m-1}$ with $\sum r_i x^i = D \bmod f_{b,c}$

for $i \leftarrow n-c$ **to** 0 **do** // strands
  if $i \geq 2c-1$ **and** $\sigma(i) \neq 1$ **then** $P_i \leftarrow h_{i-c} \oplus E_{i+c}$;
  if $\sigma(i) \neq 2$ **then** $E_i \leftarrow h_i \oplus P_{i+2c}$;
for $k \leftarrow 2c-1$ **to** $n-c$ **do** // third values
  if $\sigma(k) = 1$ **then** $P_k \leftarrow E_{k-c} \oplus E_k$;
for $k \leftarrow 0$ **to** $m-1$ **do** // values U_k
  if $k \leq c-2$ **then**
    $U_k \leftarrow E_k \oplus h_{k+b+c}$
  else if $k = c-1$ **then**
    $U_k \leftarrow E_{c-1}$
  else if $k \leq 2c-2$ **then**
    $U_k \leftarrow U_{k-c} \oplus E_k$
  else if $k \leq b+c-2$ **then**
    $U_k \leftarrow P_k$
  else $U_k \leftarrow h_{k-c}$;
for $i \leftarrow 0$ **to** $m-1$ **do** // outputs
  if $i \geq b$ **then** $r_i \leftarrow (d_i \oplus U_i) \oplus U_{i-b}$;
  else $r_i \leftarrow d_i \oplus U_i$;

Theorem 13 (Chain construction)**.** *For all $b > c \geq 1$, Algorithm 2 computes $D \bmod f_{b,c}$ using exactly $3m-3-N(b,c)$ XOR gates, where u ranges over the positive integers and*

$$N(b,c) = \#\{u \leq b+c-1 : \lfloor u/c \rfloor \equiv 2 \pmod 3\}$$
$$= c\left\lfloor \frac{b+c}{3c} \right\rfloor + \max\bigl(0,\ ((b+c) \bmod 3c) - 2c\bigr).$$

Consequently,

$$\mathrm{Red}(f_{b,c}) \leq \min\bigl\{3m-c-3,\ 3m-3-N(b,c)\bigr\}.$$

The second bound is larger than the first when $b < 2c$, equal to it when $2c \leq b \leq 4c$, and smaller by $N(b,c)-c \geq 1$ when $b \geq 4c+1$. In all cases $3m-3-N(b,c) \leq \frac{8}{3}m+c-3$.

Proof. Availability of operands. In the first loop, P_i with $\sigma(i) \neq 1$ uses E_{i+c}. Either $i+c > n-c$, and the value is free by Lemma 12, or $\sigma(i+c) = \sigma(i)+1 \in \{0,1\}$ and E_{i+c} was computed at the earlier step $i+c$. Likewise, E_i with $\sigma(i) \neq 2$ uses P_{i+2c}, which is either free or has $\sigma(i+2c) = \sigma(i)+2 \in \{0,2\}$ and index at least $2c-1$, so it was computed at step $i+2c$. In the second loop, $\sigma(k) = 1$ gives $\sigma(k-c) = 0$, so both E_{k-c} and E_k are available. The values E_k with $k \leq 2c-2$ used in the third loop have $\sigma(k) \in \{0,1\}$.

Correctness. By Lemma 12, every gate of the first two loops computes the value it names. For the values U_k we use (7). If

TABLE 3. Reduction program for $f_{6,5} = x^{16} + x^{11} + x^6 + x^5 + 1$ (Example 11).

Step	Instructions	XORs
Pairs	$Z_i = h_i \oplus h_{i+11}$, $i = 0, \ldots, 3$	4
U	$U_i = Z_i \oplus h_{i+5}$ $(i \le 3)$; $U_4 = h_4 \oplus h_9$	5
	$U_i = Z_{i-5} \oplus h_{i+5}$ $(5 \le i \le 8)$; $U_9 = h_4 \oplus h_{14}$	5
	$U_i = h_{i-5}$ for $10 \le i \le 15$	0
Outputs	$r_i = d_i \oplus U_i$ $(0 \le i \le 5)$; $r_i = d_i \oplus U_i \oplus U_{i-6}$ $(6 \le i \le 15)$	26
Total		$40 = 3m - c - 3$

$k \le c-2$, then $U_k = C_k + C_{k+c} + h_{k+b+c} = E_k + h_{k+b+c}$. At $k = c-1$, $h_{m-1} = 0$ and $U_{c-1} = E_{c-1}$. If $c \le k \le 2c-2$, then

$$\begin{aligned} U_k &= C_{k-c} + C_{k+c} + h_{k+b} \\ &= (E_{k-c} + h_{k+b}) + E_k = U_{k-c} + E_k. \end{aligned}$$

If $2c-1 \le k \le b+c-2$, then $k+b > n$ and $U_k = P_k$; for larger k, $U_k = C_{k-c} = h_{k-c}$ as in Algorithm 1. The last loop is (8).

XOR count. The values P_k with $2c-1 \le k \le n-c$ cost one gate each, in the first or in the second loop, which gives $n - 3c + 2 = b - c$ gates. The values E_t with $0 \le t \le n-c$ and $\sigma(t) \ne 2$ cost one gate each; with $u = t+1$, the excluded indices are exactly those counted by $N(b,c)$, so this gives $b + c - 1 - N(b,c)$ gates. The values U_k cost $2(c-1)$ gates, and the outputs cost $m + 2c$, as in Theorem 8. The total is

$$(b-c)+(b+c-1-N)+2c-2+m+2c = 3m-3-N(b,c).$$

The count $N(b,c)$. The integers u with $\lfloor u/c \rfloor \equiv 2 \pmod 3$ form the blocks $[2c, 3c-1]$, $[5c, 6c-1]$, and so on, of c integers each. Writing $b + c = 3c\eta + \xi$ with $0 \le \xi < 3c$, the range $1 \le u \le b + c - 1$ contains η complete blocks and $\max(0, \xi - 2c)$ elements of the next one. If $b < 2c$, then $b + c - 1 < 3c - 1$ and $N = b - c < c$. If $2c \le b \le 4c$, the first block is complete and the second is not reached, so $N = c$. If $b \ge 4c+1$, then $b + c - 1 \ge 5c$ and $N \ge c + 1$. Finally, $N \ge (b+c-2c)/3 = (b-c)/3$, and $b - c = m - 3c$ gives $3m - 3 - N \le \frac{8}{3}m + c - 3$. □

With a schoolbook multiplier, Theorem 13 gives $m^2 + m - 2 - N(b,c)$ XOR gates. For $c = 1$ the bound becomes $3m - 3 - \lfloor (m-1)/3 \rfloor \approx \frac{8}{3}m$, so members with small c, which were the most expensive under Theorem 8, now reach the same asymptotic cost as members with c close to $m/3$. For a fixed degree, the better of the two bounds is close to $\frac{8}{3}m$ at both ends of the range of c and largest, about $\frac{17}{6}m$, near $b = 4c$. These are statements about circuits for sequences of parameters. Whether such regimes contain irreducible members of arbitrarily large degree is a separate question, which we do not settle: for members with $\gcd(b,c) = 1$, the census of Section IX provides finite evidence only.

We verified Algorithm 2 symbolically for all 201 117 members with $m \le 1100$: it computes $D \bmod f_{b,c}$ exactly and uses exactly $3m - 3 - N(b,c)$ gates. It improves on Theorem 8 for 100 284 of them. The construction also explains several entries of Table 5. For $f_{6,1}$ it gives 19 gates, which is optimal, and for $f_{5,1}$ it gives 16, the optimum of Table 2. For $f_{7,1}$, $f_{8,1}$ and $f_{9,1}$ it gives 22, 24 and 27 gates, the best values that the SAT solver found. For members with $2c \le b \le 4c$, such as $f_{5,2}$, the SAT solver still finds one gate less, so some slack remains in that range.

Example 14. For $f_{8,1} = x^{10} + x^9 + x^8 + x + 1$ we have $m = 10$, $n = 8$, $c = 1$ and $\sigma(t) = (t+1) \bmod 3$. Algorithm 2 computes

$$\begin{aligned} P_7 &= h_6 \oplus h_8, & E_6 &= h_6 \oplus h_7, \\ P_5 &= h_4 \oplus E_6, & E_5 &= h_5 \oplus P_7, \\ P_4 &= h_3 \oplus E_5, & E_3 &= h_3 \oplus P_5, \\ P_2 &= h_1 \oplus E_3, & E_2 &= h_2 \oplus P_4, \\ P_1 &= h_0 \oplus E_2, & E_0 &= h_0 \oplus P_2, \\ P_3 &= E_2 \oplus E_3, & P_6 &= E_5 \oplus E_6, \end{aligned}$$

and then $U_0 = E_0$, $U_k = P_k$ for $1 \le k \le 7$, $U_8 = h_7$ and $U_9 = h_8$. These are 12 gates, against 14 in Algorithm 1 (six for the comb sums and eight for the values U_k). With the 12 output gates, the total is $24 = 3m - 3 - N(8,1)$, compared with $3m - c - 3 = 26$.

The gain has a price in depth. Along a strand, each step of $3c$ positions adds two levels instead of one.

Proposition 15 (Depth of the chain construction)**.** *The XOR depth of Algorithm 2 is at most* $2\lfloor (n-c)/(3c) \rfloor + 6$.

Proof. By induction from the top, the value E_t has depth at most $2\lfloor (n-c-t)/(3c) \rfloor + 2$, since E_t depends on P_{t+2c}, which depends on E_{t+3c}. Hence every value E_t has depth at most $\Delta = 2\lfloor (n-c)/(3c) \rfloor + 2$, every value P_k and every U_k with $k \le c-2$ has depth at most $\Delta + 1$, and every U_k with $c \le k \le 2c-2$ has depth at most $\Delta + 2$. The outputs add two more levels. □

In our verification up to $m = 1100$, the depth never exceeded $2\lfloor (n-c)/(3c) \rfloor + 5$. For the NIST-degree member $x^{571} + x^{549} + x^{527} + x^{22} + 1$, Algorithm 2 uses 1534 gates and has depth 20, against 1688 gates and depth 11 for Algorithm 1. The two constructions therefore offer a trade-off between area and delay. The word-level formulation and the low-depth variant of the next section are based on Algorithm 1.

The construction is also optimal in a restricted sense. We modeled one chain as a weighted MaxSAT problem whose

programs may use the comb sums, the values E and P, the shared pairs Z_i, and every identity of Lemma 12 and Theorem 7. For every chain length up to 1100, the optimum equals the cost of Algorithm 2, except for the shortest chains, with three indices, where the shared pairs of Algorithm 1 save one gate; such chains occur only when $b < 2c$. Shorter programs, such as the one for $f_{5,2}$, must therefore use sums that are not of the forms considered here.

VII. WORD-LEVEL FORMULATION AND A LOW-DEPTH VARIANT

The closed-form quotient of Theorem 7 can also be expressed using shifts and XOR operations on bit vectors. This formulation is suitable for software implementations, where several polynomial coefficients can be processed simultaneously. Throughout the section we work with Algorithm 1; the chain construction of Section VI has a longer critical path, and we do not pursue a word-level form of it.

We represent a polynomial by a bit vector whose bit i is the coefficient of x^i. The notation $A \gg k$ denotes a right shift by k positions, corresponding to $\lfloor A/x^k \rfloor$, while $A \ll k$ denotes multiplication by x^k.

Corollary 16 (Word-level reduction)**.** *With L and H as defined in* (1)*, let*

$$\begin{aligned} C &= \textstyle\sum_{\ell \geq 0} (H \gg 3c\ell), \\ Q &= C \oplus (C \gg c) \oplus (H \gg (b+c)), \\ U &= Q \oplus (Q \ll c). \end{aligned}$$

Then

$$D \bmod f_{b,c} = \big(L \oplus U \oplus (U \ll b)\big) \bmod x^m.$$

The vector C can be computed by initializing $C \leftarrow H$ and applying

$$C \leftarrow C \oplus (C \gg 3c \cdot 2^j)$$

for successive integers $j \geq 0$ such that $3c \cdot 2^j \leq n$.

Proof. The coefficient of x^t in $\sum_{\ell \geq 0}(H \gg 3c\ell)$ is the comb sum C_t. The expression for Q follows from (6), while $U = (1 + x^c)Q$ and the expression for the remainder follow from (8).

After doubling step j, each coefficient of C contains the sum of the corresponding 2^{j+1} terms, or all available terms if fewer remain. Thus, the complete comb sums are obtained once $3c \cdot 2^{j+1} > n$. □

The vector formulation evaluates all comb sums through a sequence of shifts and XOR operations. The correction term $H \gg (b+c)$ can have nonzero coefficients only at positions $j \leq c - 2$. In the final expression, reduction modulo x^m amounts to discarding bits above position $m - 1$.

The doubling procedure avoids computing each shifted copy of H separately. At each step, the number of terms accumulated in each position doubles. For example, with $b = 7$ and $c = 1$, we have $n = 7$ and stride 3. The first step combines h_t and h_{t+3}, while the second includes h_{t+6} and h_{t+9}, with out-of-range coefficients taken as zero.

Let

$$\nu = \left\lceil \log_2 \left(\frac{n+1}{3c} \right) \right\rceil,$$

with $\nu = 0$ when $3c > n$. The reduction requires $\nu + 5$ shift-and-XOR operations on bit vectors: ν for the comb sums, two for Q, one for U, and two for the remainder. Hence the number of vector operations grows logarithmically with m/c.

This count assumes operations on complete bit vectors and does not directly represent the number of machine instructions. The latter depends on the processor architecture, register width and representation of the field elements.

When $b \leq 4c + 1$, we have $\nu \leq 1$, and the reduction requires at most six shift-and-XOR operations. This includes the selected members of degrees 163 and 571 discussed in Section IX. The formulation can be used after carry-less polynomial multiplication, for example with `PCLMULQDQ`, or in bitsliced implementations.

Zhou and Zhang [31] recently gave a general reduction for every monic binary modulus, in which the feedback recurrence is inverted by a product of Frobenius powers, with exactly $\lceil \log_2(m/\Delta_{\min}) \rceil$ sequential feedback stages, where $\Delta_{\min}$ is the distance from m to the highest nonleading exponent. For $f_{b,c}$ this distance is $m - (b + c) = c$. The doubling steps of Corollary 16 number $\nu = \lceil \log_2((n+1)/(3c)) \rceil$, one or two fewer for every member with $m \leq 1100$, because the comb sums obey $C_j + C_{j+3c} = h_j$, a recurrence with a single feedback term at distance $3c$ that comes from the sparse multiple of Proposition 2(ii). Their work counts word operations and does not count XOR gates; the contribution here is specific to the second family: the closed form, the exact gate counts of Theorems 8 and 13 and the depth bounds.

The same doubling procedure can be applied at the gate level to reduce the depth of the comb-sum computation.

Proposition 17 (Low-depth variant)**.** *Replacing the sequential comb sums in Algorithm 1 with the doubling procedure of Corollary 16 gives a reduction circuit with at most*

$$3m - c - 3 + \sum_{j=1}^{\nu-1} \max\big(0,\ n + 1 - 3c \cdot 2^j\big)$$

XOR gates and depth at most $\nu + 4$.

Proof. At doubling step j, an XOR gate is required for each position t satisfying $t + 3c \cdot 2^j \leq n$. The number of gates at that step is therefore

$$\max(0, n + 1 - 3c \cdot 2^j).$$

Each step adds one level to the circuit depth.

The first step, $j = 0$, requires $n + 1 - 3c = b - c - 1$ gates, matching the comb-sum cost in Algorithm 1. The remaining steps account for the additional gates in the stated bound.

The subsequent stages are unchanged and contribute at most four levels, as established in Proposition 9. Thus, the overall depth is at most $\nu + 4$. □

The two implementations provide different trade-offs between gate count and circuit depth. In Algorithm 1, each comb sum is computed sequentially from the preceding one. A sum containing d terms therefore requires a chain of $d-1$ dependent XOR gates.

The doubling procedure combines partial sums in parallel, reducing the depth to $\lceil \log_2 d \rceil$, although additional gates may be required. When $b \leq 4c+1$, each comb sum contains at most two terms, and the two constructions coincide. The low-depth variant is therefore most relevant when c is small relative to m.

We verified the low-depth variant symbolically for all members with $m \leq 1100$, including its gate count and depth bound. As an illustration, consider

$$f = x^{571} + x^{549} + x^{527} + x^{22} + 1,$$

where $b = 527$, $c = 22$, $n = 569$ and the stride is 66.

The comb sums contain up to nine terms, giving $\nu = 4$. The sequential construction of Algorithm 1 has depth 11 and uses 1688 XOR gates. The doubling variant reduces the depth to 7 but requires 2474 gates.

Thus, the reduction of four depth levels comes at the cost of 786 additional XOR gates, an increase of approximately 47%. The choice between the two constructions depends on the area and delay requirements of the implementation.

VIII. COMPARISON WITH THE GREEDY COUNT

Banegas' second family was originally identified using Conta-XOR [22]. We compare the reduction costs obtained from the closed-form construction with those produced by this heuristic. The analysis follows the classification introduced in Definition 3: members with $b/c \in \{\frac{3}{2}, 2, 3, 4, 5\}$ are considered special, while all others are generic.

Observation 18. *We evaluated* $\mathrm{CX}(f_{b,c})$ *for all* 3 674 *parameter pairs with* $5 \leq m \leq 150$. *For each pair, we also performed* 20 *additional greedy runs with randomized tie-breaking, following the approach of [27].*

Among the 3 541 *generic members, we observed the following:*

- *(a) The closed-form construction was never more expensive than either greedy result.*
- *(b) For all* 879 *members with* $b < 2c$, $\mathrm{CX}(f) = 2m + 3c - 3$. *The closed-form construction therefore used exactly* $2c - b$ *fewer XOR gates.*
- *(c) For all* 1 312 *members satisfying* $2c < b < 6c$, *the greedy count was* $\mathrm{CX}(f) = 3m - c - 3$, *equal to the closed-form count.*
- *(d) For* $b \geq 6c$, *the greedy count exceeded the closed-form count in* 392 *of the* 1 350 *cases, with differences of up to* 112 *XOR gates.*

Randomized tie-breaking improved the greedy count for 212 *generic members, all with* $b \geq 6c$, *by at most* 12 *gates. In only one of these cases did the improvement eliminate the difference relative to the closed-form construction. These figures depend on the random choices; an independent rerun gave* 213 *members and at most* 13 *gates, again all with* $b \geq 6c$ *and with a single case closing the gap.*

For special members, the greedy counts follow the strip bound (2), *with the values of* $R_1(G)$ *and* $R_2(G)$ *that the greedy algorithm finds for each base polynomial. For* $f_{2,1}$ *and* $f_{3,1}$ *these are the optimal values of Table 2; for* $f_{4,1}$, $f_{5,1}$ *and* $f_{3,2}$ *the lexicographic greedy algorithm finds programs one or two gates longer (*$R_1 = 16$ *for* $f_{4,1}$, *and* $R_2 = 17$, $R_1 = 19$ *for* $f_{5,1}$ *and* $f_{3,2}$*). For example, when* $b = 3c$, *the observed cost was* $\frac{12}{5}m - 2$ *XOR gates. Since the reduction matrix of* f *is block diagonal over the strips (Proposition 4), the greedy algorithm never mixes strips and effectively reduces modulo* G *once per strip.*

The results indicate that randomized tie-breaking has limited influence on the differences observed between the two constructions. This suggests that the choice of intermediate expressions, rather than tie-breaking alone, may account for part of the difference. The experiments do not, however, establish that the closed-form counts are optimal.

Figure 2 shows $\mathrm{CX}(f) - (3m - c - 3)$ as a function of b/c, using a logarithmic horizontal scale. The figure includes all generic members with $m \leq 150$. Every difference is nonnegative, and three regions can be distinguished.

For $b/c < 2$, the difference is exactly

$$2c - b = c(2 - b/c).$$

At a fixed ratio, the difference increases with c and therefore with the field degree. The largest difference in this region is 48 gates, obtained for $b = c + 1$ and $c = 49$.

For $2 < b/c < 6$, both methods produce programs with the same XOR count. When $b/c \geq 6$, the greedy algorithm produces larger counts for some members, particularly when c is small. In these cases, the comb sums contain more terms, and the ordering of intermediate XOR operations becomes increasingly relevant. The curves extending toward larger ratios correspond to fixed small values of c, such as $c = 1$ and $c = 2$, with increasing m.

The same regularities hold beyond the range of Observation 18. We ran the greedy algorithm, with lexicographic tie-breaking, on all 11 175 members with $151 \leq m \leq 300$ and on all members of degrees 500, 800 and 1100. For every generic member, the closed form was never more expensive; the greedy count was exactly $2m + 3c - 3$ when $b < 2c$ and exactly $3m - c - 3$ when $2c < b < 6c$. For $b \geq 6c$, the greedy algorithm needed more gates in 598 of the 4 163 members with $151 \leq m \leq 300$ and in 36 of the 299 members of the three larger degrees, by up to 248 and 994 XOR gates, respectively. The fraction of such members decreases as m grows, but the excess, when it occurs, becomes larger.

The same comparison can be made for the polynomials reported in Banegas' thesis. Table 4 lists the irreducible members of the second family at the NIST field degrees, including those reported in [22, Tab. 17].

For the listed members with $b > 2c$, the greedy counts coincide with $3m - c - 3$. When $b < 2c$, the closed-form

construction requires fewer gates. The two listed members with $b \geq 4c+1$ are improved by the chain construction: the degree-233 member with $c = 16$ needs 632 gates instead of 680, and the degree-571 member with $c = 22$ needs 1534 instead of 1688. Neither changes the best member of its degree.

For example,

$$f = x^{163} + x^{117} + x^{71} + x^{46} + 1$$

has $b = 71$ and $c = 46$. Its greedy count is 461, whereas the closed-form construction requires 440 gates, a reduction of $21 = 2 \cdot 46 - 71$ XOR gates.

We also compared the closed-form construction with the heuristic of Boyar and Peralta [24], which can exploit cancellations between intermediate expressions.

For the generic degree-9 and degree-10 members, $f_{5,2}$ and $f_{4,3}$, the best of 10 runs produced programs with 22 and 24 XOR gates, respectively. Both counts match $3m - c - 3$, whereas Conta-XOR gave 22 and 26.

For the degree-7 special member $f_{3,2}$, both the closed-form construction and the Boyar and Peralta heuristic required 16 gates, compared with 17 for Conta-XOR.

The closed-form derivation also uses cancellations, notably the elimination of C_k in $U_k = q_k + q_{k-c}$. Unlike the heuristics, however, Algorithm 1 incorporates these cancellations directly into its expressions, without searching for intermediate XOR combinations.

Exact minima for small members

To find out how far the closed form is from optimal, we computed exact minima with a SAT solver, using the encoding of [33]. The search is made feasible by the following reduction to the high inputs.

Lemma 19 (Elimination of the low inputs)**.** *Let $f = f_{b,c}$ and let R' be the reduction matrix of f restricted to the high inputs $h_0, \ldots, h_n$, so that $r_i = d_i + \ell_i(h)$ with ℓ_i the ith row of R'. Then every ℓ_i is nonzero, and in the model of Section II,*

$$\mathrm{Red}(f) = m + \mathrm{Red}(R').$$

Proof. By (8), $\ell_i = U_i + U_{i-b}$, with $U_{i-b} = 0$ for $i < b$. By (7), U_k contains h_k for $0 \leq k < c$ and h_{k-c} for $k \geq c$, and all its other terms have larger indices. Let $\lambda(k)$ be this smallest index; it is at most $m - 1 - c \leq n$, so the term is present. For $i < b$, $\ell_i = U_i \neq 0$. For $i \geq b$, $\lambda(i-b) \leq i-b < i-c = \lambda(i)$, so $h_{\lambda(i-b)}$ occurs in U_{i-b} but not in U_i, and $\ell_i \neq 0$.

A circuit for R' followed by one gate $r_i = d_i \oplus \ell_i$ per output gives $\mathrm{Red}(f) \leq m + \mathrm{Red}(R')$. Conversely, take any circuit for the reduction map. Each d_i occurs only in r_i, and $r_i \neq d_i$ because $\ell_i \neq 0$, so some gate receives d_i directly. Setting $d_i = 0$ turns that gate into a copy of its other operand (or into the constant 0), so it can be removed without changing the outputs for $d_i = 0$. Doing this for $i = 0, \ldots, m-1$ removes at least m gates and leaves a circuit for R'. Hence $\mathrm{Red}(f) \geq m + \mathrm{Red}(R')$. □

The hypothesis $\ell_i \neq 0$ matters: for a modulus whose reduction matrix had a zero high row, the corresponding output would cost no gate. The programs were found with CaDiCaL 1.5.3 through PySAT. Each value stated as optimal in Tables 2 and 5 is backed by the program itself, checked symbolically against the reduction matrix, and by a DRAT certificate of unsatisfiability for one gate less, produced by CaDiCaL 3.0.1 and checked with drat-trim; a program with fewer gates can always be padded with unused gates, so this rules out every shorter program. Values without such a certificate are marked as upper bounds. The programs, the CNF instances and the certificates are in the repository given at the end of the paper. Table 5 summarizes the results. For the base polynomials of Definition 3, the values of Table 2 are optimal. For every small generic member we could solve, however, the optimum lies at least one gate below the closed form, and for $c = 1$ the known gap grows with b. Neither the greedy algorithm, with up to 500 randomized runs, nor the heuristic of Boyar and Peralta finds these programs. The bound of Theorem 8 is therefore not optimal in general. The chain construction of Section VI attains the optimum for $f_{6,1}$ and $f_{5,1}$, and the best values found by the SAT solver for $f_{7,1}$, $f_{8,1}$ and $f_{9,1}$. For $f_{5,2}$ and $f_{7,2}$, which satisfy $2c \leq b \leq 4c$, a gap of one gate remains.

IX. COMPARISON WITH THE FIRST FAMILY AND WITH OTHER POLYNOMIALS

A. PARAMETER SELECTION

The bound of Theorem 8 decreases as c increases, while the bound of Theorem 13, which matters when $b \geq 4c+1$, grows roughly as $\frac{8}{3}m + \frac{1}{3}c$. For a fixed degree, the better of the two is therefore smallest at the two ends of the range of c, where it approaches $\frac{8}{3}m$, and largest near $b = 4c$, where it is about $\frac{17}{6}m$. Special members can do better (Table 2), but they exist only in degrees divisible by 4, 5, 6 or 7, and they are irreducible only in the few degrees described by Proposition 20. For a given field degree, the best choice is the irreducible member that minimizes $\min\{3m - c - 3,\ 3m - 3 - N(b,c)\}$, and it usually has c either close to $m/3$ or very small.

For Banegas' first family, the corresponding bound is $3m - 2$, independently of b and c, except for the case $m = 5c$. Thus, when comparing the general bounds at the same degree, the second family offers a reduction of at least $c + 1$ XOR gates.

B. IRREDUCIBILITY CENSUS

We tested all members of both families, as well as all trinomials, for irreducibility in degrees $5 \leq m \leq 1100$, using Rabin's test [34]. The results are summarized in Table 6.

Whenever an irreducible trinomial exists, its known reduction cost of $2m - 2$ XOR gates is lower than the general bounds considered here for the two pentanomial families; the only exceptions are three special members, discussed in Section IX-D. We therefore focus on the 502 degrees in the evaluated range for which no irreducible trinomial exists. All costs compared in this section are costs of explicit

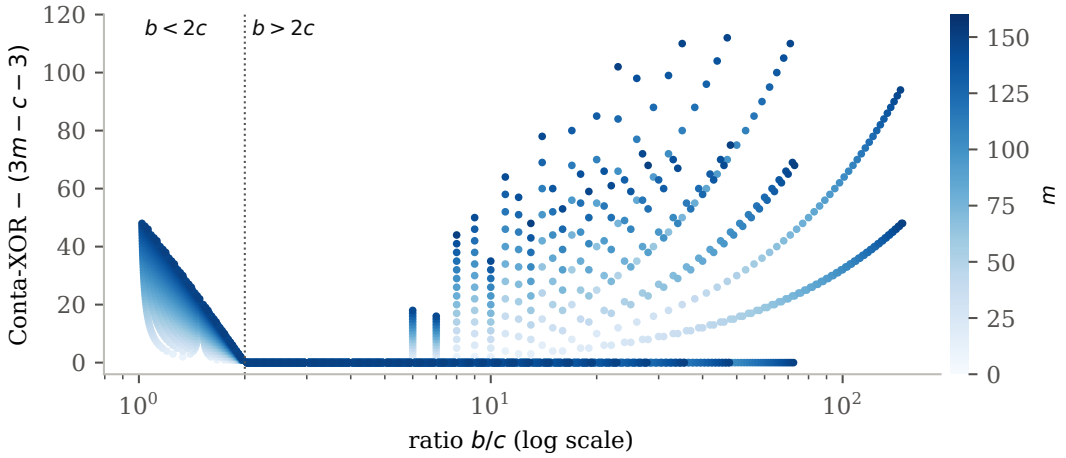


FIGURE 2. Difference between the Conta-XOR count and the closed-form count, $\mathrm{CX}(f) - (3m - c - 3)$, for the 3 541 generic members with $m \leq 150$. Color indicates the degree m, and the dotted line marks $b/c = 2$. The difference is nonnegative throughout the evaluated range. For $b/c < 2$, it equals $2c - b$; for $2 < b/c < 6$, it is zero. Larger differences occur for some members with $b/c \geq 6$.

TABLE 4. Irreducible members of the second family at the NIST field degrees. The label "thesis" identifies values reported in [22, Tab. 17]. The depth refers to Algorithm 1; the last column is the cost of the chain construction (Theorem 13).

m	f	b	c	dXOR	CX	$3m - c - 3$	depth	$3m - 3 - N$
163	$x^{163} + x^{117} + x^{71} + x^{46} + 1$	71	46	649	461	**440**	4	461
233	$x^{233} + x^{158} + x^{83} + x^{75} + 1$	83	75	866	688	**621**	4	688
233	$x^{233} + x^{217} + x^{201} + x^{16} + 1$	201	16	1687	680	680	7	**632**
409	$x^{409} + x^{294} + x^{179} + x^{115} + 1$	179	115	1642	1160	**1109**	4	1160
409	$x^{409} + x^{337} + x^{265} + x^{72} + 1$	265	72	1823	1152	1152	4	1152
571	$x^{571} + x^{389} + x^{207} + x^{182} + 1$	207	182	2145	1685 (thesis)	**1528**	4	1685
571	$x^{571} + x^{465} + x^{359} + x^{106} + 1$	359	106	2531	1604 (thesis)	1604	4	1604
571	$x^{571} + x^{549} + x^{527} + x^{22} + 1$	527	22	6297	1688 (thesis)	1688	11	**1534**

TABLE 5. Exact minima $\mathrm{Red}(f)$ for small members, computed with a SAT solver, compared with the closed form, with the best bound of Theorems 8 and 13, and with the best heuristic result (greedy with randomized ties, or Boyar and Peralta). An entry "$\leq v$" means that a program with v gates exists and that smaller values were not decided.

f	m	$3m - c - 3$	best bound	heuristic	$\mathrm{Red}(f)$
$f_{2,1}$ (special)	4	8	8	6	6
$f_{3,1}$ (special)	5	11	11	10	10
$f_{4,1}$ (special)	6	14	14	14	14
$f_{5,1}$ (special)	7	17	16	16	16
$f_{3,2}$ (special)	7	16	16	16	16
$f_{6,1}$	8	20	19	20	19
$f_{5,2}$	9	22	22	22	21
$f_{7,1}$	9	23	22	23	≤ 22
$f_{8,1}$	10	26	24	26	≤ 24
$f_{7,2}$	11	28	28	28	≤ 27
$f_{9,1}$	11	29	27	29	≤ 27

constructions, or greedy counts where stated, and hence upper bounds on $\mathrm{Red}(f)$. A comparison of upper bounds does not by itself order the minima $\mathrm{Red}(f)$ of two families; it compares the best circuits known for each.

For these degrees, Fig. 3 compares the reduction cost per bit, defined as the number of XOR gates divided by m, for the lowest-cost irreducible members identified in each family.

For the first family, the general bound gives $3 - 2/m$ gates per bit, apart from the case $b = 2c$. This value approaches 3 as m increases.

For the second family, the corresponding expression is the smaller of

$$3 - \frac{c+3}{m} \quad \text{and} \quad 3 - \frac{N(b,c)+3}{m}.$$

The first approaches $\frac{8}{3}$ when c is close to $m/3$, and the second when c is small. Across the evaluated degrees, the mean cost per bit of the best generic member is 2.70, against 2.77 with Theorem 8 alone. The chain construction lowers the cost of the best member of the second family in 379 of the 720 degrees, 160 of them without irreducible trinomials, by up to 11.1%. For example, at $m = 975$ the member with $c = 1$ needs 2598 XOR gates instead of 2921.

Among the 405 degrees containing irreducible members of both families, the second family has a lower reduction bound in all of them, with a mean saving of 9.7%, against 7.3% with Theorem 8 alone. Two of these degrees depend on special members. At $m = 5$, the special member $f_{3,1}$ needs 10 XOR

TABLE 6. Irreducibility census for $5 \le m \le 1100$ (1 096 degrees). The comparison uses the better of Theorems 8 and 13 for generic members of the second family and the strip bound (2) with Table 2 for special members, and $3m-2$ or $\frac{12}{5}m-1$ for the first family [21].

irreducible members of the second family	1 508
degrees with at least one member: second family / first family	720 / 502
degrees with members of both families	405
of which the second family has the lower bound	405 (mean saving 9.7%)
degrees in which Theorem 13 improves the best member	379
degrees without irreducible trinomials	502
covered by the second family / by the first family	288 / 229
covered by the second family only	112
covered by neither	161
mean cost per bit of the best generic member of the second family	2.70
the same with Theorem 8 only	2.77

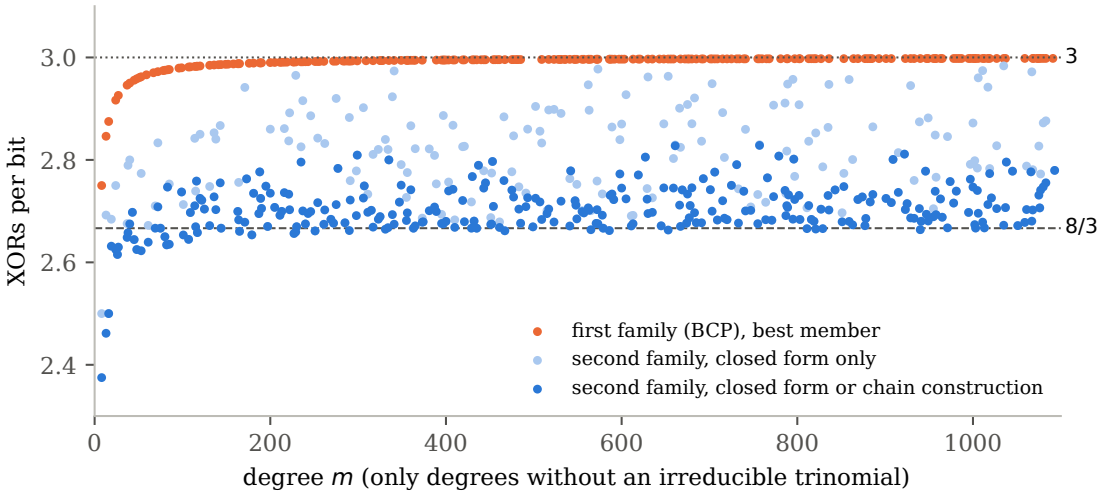


FIGURE 3. Reduction cost per bit for degrees $m \le 1100$ without irreducible trinomials. The figure compares the lowest reported bounds for members of Banegas' first family (BCP) and the second family, with Theorem 8 alone and with the better of Theorems 8 and 13. The dashed line indicates the asymptotic limit $\frac{8}{3}$ of both general bounds for the second family.

gates, against 11 for the first family. At $m = 155$, the first-family member with $b = 2c$ requires $\frac{12}{5}m - 1 = 371$ XOR gates, and the best generic member of the second family needs 411. However, the special member

$$f_{93,31} = x^{155} + x^{124} + x^{93} + x^{31} + 1 = f_{3,1}(x^{31})$$

is irreducible, and the strip bound gives $\frac{12}{5}m - 2 = 370$ XOR gates.

The special members behind these two degrees are examined in Section IX-D.

The second family also covers more field degrees. Among the 502 degrees without irreducible trinomials, 176 contain members of both families, 112 contain only members of the second family, and 53 contain only members of the first. The remaining 161 degrees contain no irreducible member of either family.

Degrees satisfying $m \equiv 0 \pmod 8$ are of particular interest because they admit no irreducible trinomials. The second family contains irreducible members in 25 such degrees, including $x^{16} + x^{11} + x^6 + x^5 + 1$ (Example 11), $x^{48}+x^{33}+x^{18}+x^{15}+1$ and $x^{144}+x^{99}+x^{54}+x^{45}+1$, whose reduction costs are 40, 126 and 384 XOR gates, respectively.

C. COMPARISON WITH OTHER PENTANOMIAL FAMILIES

Banegas' first family is not the only family of pentanomials proposed for degrees without irreducible trinomials. We repeated the census for type I pentanomials $x^m + x^{n+1} + x^n + x + 1$ and type II pentanomials $x^m + x^{n+2} + x^{n+1} + x^n + 1$, with $n \le m/2 - 1$ [17], and for the class $x^m + x^{m-s} + x^{m-2s} + x^{m-3s} + 1$ of [18]. Each of these degrees also has an irreducible pentanomial with $k_3 \le m/2$, whose reduction costs at most $4m - 4$ XOR gates (Table 1). For type I members we use the bound $3m - 2$ of Table 1. For type II members we use the greedy count, the same measure as for the NIST pentanomials in Table 9, minimized over the irreducible members of each degree. The class of [18] has no irreducible member in the degrees considered here: all its irreducible members with $m \le 1100$ lie in degrees that have an irreducible trinomial.

Table 7 summarizes the comparison over the 502 degrees without irreducible trinomials. In each of the 288 degrees

where the second family has an irreducible member, its best member has a lower reduction bound than the best member of every other family, by 5.6% to 33.2%, with a mean of 11.7%. Type I pentanomials cover more degrees than the second family, 361 against 288, but at the cost of the first family, about 3 XOR gates per bit. Taken together, the four families cover 484 of the 502 degrees; in 15 of them, for example $m = 69$ and $m = 306$, the second family is the only one.

D. IRREDUCIBLE SPECIAL MEMBERS

For members with $\gamma > 1$, irreducibility can be decided from the base polynomial alone. This gives a complete description of the irreducible special members.

Proposition 20 (Irreducible special members)**.** *Let $f = f_{b,c} = G(x^\gamma)$ as in Definition 3, with $\kappa = \deg G$. Then f is irreducible over $\mathbb{F}_2$ if and only if G is irreducible and, denoting by e the order of G, every prime divisor of γ divides e and $\gcd\big(\gamma, (2^\kappa - 1)/e\big) = 1$. In particular, a special member is irreducible if and only if one of the following holds:*

(i) $G = f_{2,1}$ *and* γ *is a power of* 5*;*
(ii) $G = f_{3,1}$ *and* γ *is a power of* 31*;*
(iii) $G = f_{4,1}$ *and* $\gamma = 3^i 7^j$ *with* $i, j \geq 0$.

Proof. If $G = AB$ is a nontrivial factorization, then $f = A(x^\gamma)B(x^\gamma)$ is reducible. For irreducible G, the first statement is Theorem 3.35 of [1] over $\mathbb{F}_2$. That theorem has a third condition, namely that $4 \mid \gamma$ implies $4 \mid 2^\kappa - 1$; since e is odd, the first condition already forces γ to be odd, so the third one always holds.

For the special members, $f_{5,1} = (x^3 + x + 1)(x^4 + x^3 + 1)$ and $f_{3,2} = (x^3 + x + 1)(x^4 + x + 1)$ are reducible, so all their stretched copies are reducible. The polynomial $f_{2,1} = x^4 + x^3 + x^2 + x + 1$ is irreducible of order $e = 5$, and $(2^4 - 1)/5 = 3$. The polynomials $f_{3,1}$ and $f_{4,1}$ are irreducible of orders $31 = 2^5 - 1$ and $63 = 2^6 - 1$, that is, primitive, so the gcd condition is automatic. The three cases follow. □

Table 8 lists the 15 irreducible special members with $5 \leq m \leq 1100$, the range of the census, as given by Proposition 20. For each of them, it compares the closed form of Theorem 8, the strip bound (2) and the best irreducible generic member of the same degree. The strip bound improves on the closed form in 14 of the 15 cases; the exception is $f_{4,1}$ itself, for which $\gamma = 1$ and the two coincide. All these members have $b \leq 4c$, so the chain construction does not improve them. In the 8 degrees that also have an irreducible generic member, the special member is cheaper by 6.9% to 35.3% than the best generic member under Theorems 8 and 13. In the other 7 degrees, the special member is the only irreducible member of the second family. The largest savings come from the equally spaced members, $G = f_{2,1}$, whose cost is $\frac{7}{4}m - 1$.

All 15 degrees also have irreducible trinomials. In 12 of them the trinomial cost $2m - 2$ is lower; for instance, at $m = 155$ a trinomial needs 308 XOR gates, against 370 for $f_{93,31}$. In the other three degrees, $m = 20$, 100 and 500, the special member is an equally spaced pentanomial, whose cost $\frac{7}{4}m - 1$ is below $2m - 2$ for every $m > 4$: it needs 34, 174 and 874 XOR gates, against 38, 198 and 998 for the trinomial. This advantage comes from the equally spaced subfamily, whose cost is already known (Table 1), and it is a comparison with the reference cost $2m - 2$, not with every possible circuit for every trinomial.

E. COMPARISON AT NIST FIELD DEGREES

Table 9 compares the lowest reduction costs reported for the two families at the NIST field degrees with the costs of the corresponding NIST polynomials and irreducible trinomials, where available.

At degrees 163 and 571, where no irreducible trinomial exists, the second family provides lower reduction costs than both the NIST pentanomials and the first family.

For $m = 163$, the reduction cost is 440 XOR gates, compared with 571 for the NIST pentanomial and 487 for the first family. These differences correspond to reductions of 22.9% and 9.7%, respectively.

At $m = 571$, the corresponding cost is 1528 XOR gates, compared with 2003 for the NIST pentanomial and 1711 for the first family, representing reductions of 23.7% and 10.7%.

Type I members exist at degrees 163 and 283, with the same cost $3m - 2$ as the first family, and the best type II members need 665, 1021 and 2149 XOR gates at degrees 163, 283 and 571 (greedy count). No irreducible member of the second family was found at degree 283. At degrees 233 and 409, irreducible trinomials provide lower reduction costs than the pentanomials considered here.

The reduction is only one part of a field multiplier, and Table 10 puts these savings in context at degrees 163 and 571. The polynomial product does not depend on f. For it we use the schoolbook product, with $(m - 1)^2$ XOR gates, and a recursive Karatsuba multiplier [35] that, at each size, takes the cheapest of the schoolbook product, the Karatsuba split and Bernstein's refined split [36]; it needs 12 001 XOR and 6 203 AND gates for $m = 163$, and 90 703 XOR and 47 916 AND gates for $m = 571$. The absolute saving of the reduction carries over to the multiplier unchanged, but in relative terms it becomes small: about 1% against the NIST pentanomials with Karatsuba, and less with the schoolbook product. Squaring is different. The square of a polynomial over $\mathbb{F}_2$ costs no gates before the reduction, so the reduction of a polynomial with only even-degree terms is the whole cost of a squaring. Squarings dominate inversion by the method of Itoh and Tsujii [37], and a general bit-parallel squaring for low-weight moduli is given in [30]. For squaring, we compare the greedy counts of the reduction matrices restricted to the even inputs, the same measure for every polynomial: the second family saves 13.1% and 4.4% against the first family, and 29.7% and 24.6% against the NIST pentanomials, at degrees 163 and 571.

X. DISCUSSION AND CONCLUSION

This work completes the analysis of Banegas' second family of pentanomials, a problem left open in [22]. The main result

TABLE 7. Comparison with other pentanomial families over the 502 degrees $5 \le m \le 1100$ without irreducible trinomials. For each family: the cost used, the number of degrees with an irreducible member, the mean number of XOR gates per bit of the best member, and the degrees shared with the second family, where the bound of the second family is always lower.

Family	cost used	degrees	XORs per bit	shared degrees	mean saving
second family (this paper)	Theorems 8 and 13, (2)	288	2.70		
Banegas' first family	$3m-2$, $\frac{12}{5}m-1$ [21]	229	2.99	176	9.7%
type I	$3m-2$ [18]	361	2.99	204	9.5%
type II	greedy count	318	3.73	183	27.0%
pentanomial with $k_3 \le m/2$	$4m-4$ [18]	502	3.98	288	32.1%

TABLE 8. Irreducible special members with $5 \le m \le 1100$. Closed form: $3m-c-3$ (Theorem 8); strips: bound (2) with Table 2; generic: best irreducible generic member of the same degree, with the better of Theorems 8 and 13.

m	G	γ	closed form	strips	generic
5	$f_{3,1}$	1	11	**10**	none
6	$f_{4,1}$	1	14	**14**	none
18	$f_{4,1}$	3	48	**44**	none
20	$f_{2,1}$	5	52	**34**	51
42	$f_{4,1}$	7	116	**104**	none
54	$f_{4,1}$	9	150	**134**	144
100	$f_{2,1}$	25	272	**174**	269
126	$f_{4,1}$	21	354	**314**	none
155	$f_{3,1}$	31	431	**370**	411
162	$f_{4,1}$	27	456	**404**	439
294	$f_{4,1}$	49	830	**734**	none
378	$f_{4,1}$	63	1 068	**944**	1 026
486	$f_{4,1}$	81	1 374	**1 214**	1 333
500	$f_{2,1}$	125	1 372	**874**	1 342
882	$f_{4,1}$	147	2 496	**2 204**	none

TABLE 9. Reduction costs at the NIST field degrees. NIST pentanomial counts are obtained with Conta-XOR [22]; first-family bounds are from [21]; second-family bounds follow from Theorem 8; the chain construction does not change them. "None" indicates that no irreducible polynomial of the specified type exists at that degree.

m	NIST	trinomial	first family	second family
163	571	none	487	**440**
233	464 (trinomial)	**464**	697	621
283	862	none	**847**	none
409	816 (trinomial)	**816**	none	1109
571	2003	none	1711	**1528**

is a closed-form expression for the quotient in polynomial division, from which we obtain a reduction algorithm requiring at most $3m-c-3$ XOR gates. Unlike approaches based on greedy optimization, the construction follows directly from the algebraic structure of the defining polynomial and applies uniformly to all parameters $b > c \ge 1$. The closed form is not optimal, and the same algebra also shows how to improve it. Rewriting the reduction in terms of sums of pairs of comb sums splits it into c independent chains and gives a second construction with $3m-3-N(b,c)$ XOR gates, which is shorter whenever $b \ge 4c+1$. The two constructions complement each other: the first approaches $\frac{8}{3}m$ gates when c is close to $m/3$, the second when c is small, and the first has the smaller depth.

TABLE 10. XOR gates of a full multiplier and of a squaring at degrees 163 and 571, for the best member of each family and for the NIST pentanomial. Multiplier: product plus reduction, with the schoolbook product or with the recursive Karatsuba multiplier described in the text. Squaring: greedy count of the reduction of a square.

m	polynomial	reduction	schoolb.	Karatsuba	square
163	second family	**440**	**26 684**	**12 441**	**173**
	first family	487	26 731	12 488	199
	NIST	571	26 815	12 572	246
571	second family	**1 528**	**326 428**	**92 231**	**649**
	first family	1 711	326 611	92 414	679
	NIST	2 003	326 903	92 706	861

The analysis also clarifies the difference between the two families identified by Banegas. In the first family, the corresponding reduction relations involve a stride of $3b$, which exceeds the field degree. In the second family, the stride is $3c < m$, and the quotient may depend on several coefficients separated by this distance. The closed-form expression makes these dependencies explicit and allows intermediate sums to be shared across the reduction circuit. This structure is not always captured by greedy pair substitution, as shown by the comparisons with Conta-XOR.

The computational results indicate that the second family provides useful alternatives to previously studied pentanomials. Among the degrees examined, it contains irreducible members in cases where the first family does not, including degrees for which irreducible trinomials are unavailable, and wherever it has members in those degrees its reduction bounds are lower than those of the first family and of type I and type II pentanomials. At degrees 163 and 571, the proposed construction gives lower XOR counts for polynomial reduction than the corresponding NIST pentanomials and the available members of the first family. The word-level formulation and the low-depth variant further show how the algebraic result can be adapted to different implementation models. These comparisons concern reduction costs and do not, by themselves, establish an advantage in overall hardware area, energy consumption or software execution time; the best modulus depends on the architecture and on the operation [38]. In a full multiplier the saving is small in relative terms, about 1% with a Karatsuba product (Table 10); it matters most where the reduction dominates, as in squaring and therefore in inversion.

Several questions remain open. The chain construction matches the best programs found with a SAT solver for the small members with $c = 1$, but for members with $2c \le b \le 4c$, such as $f_{5,2}$, a gap of one gate remains (Table 5), and the programs that close it use sums outside the family of Lemma 12. Finding a construction that is optimal for all parameters, and lower bounds that match it, is the main open problem. For special members, the strip programs of Table 2 are optimal, but the list of Definition 3 reflects what heuristics find; optimal strip programs for larger base polynomials, such as $f_{6,1}$ and $f_{5,2}$, would extend the strip bound (2) to further ratios b/c. For members with $\gamma > 1$, irreducibility reduces to that of the base polynomial, and for special members it is completely described by Proposition 20. For members with $\gcd(b, c) = 1$, however, irreducibility is known only from the computational census, and algebraic conditions remain to be found. Finally, jointly optimizing polynomial multiplication and reduction, as in Mastrovito-type multipliers [18], [39], and evaluating the resulting circuits under realistic area and delay constraints would help determine their practical advantages.

The generators for both reduction algorithms, the symbolic verification code, the SAT models, the optimal programs with their DRAT certificates, the irreducibility census and the scripts used to produce the experimental results are available at https://github.com/rfcustodio/banegas-second-family-verification. These materials support independent verification and reproduction of the reported results.

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